



Connes Fusion

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Following Chapter 3 of [Unitary Quantum Symmetries Lite](#) from Penneys, Ferrer, and Kawagoe we want to try to first establish what Connes fusion is. Big picture: just like when we compose (fuse) group representations on Hilbert spaces, we end up in the tensor product Hilbert space; Connes fusion is the equivalent for modules over an algebra.

This text gives three equivalent definitions. For now let's restrict ourselves to the first one.

Definition: Connes Fusion.

Given a right B -module H_B and a left B -module ${}_B K$. We define the **Connes Fusion** relative tensor product Hilbert space $H \boxtimes_B K$ can be defined using the relative tensor product for C^* -Hilbert modules. $\text{Hom}(L^2 B_B \rightarrow H_B) \boxtimes_B K$ with $\mathbb{C}$ -valued inner product $\langle f_1 \boxtimes \eta_1 \mid f_2 \boxtimes \eta_2 \rangle := \langle f_2 \mid f_1 \rangle_B \langle \eta_1 \mid \eta_2 \rangle_K$ where we have $\langle f_2 \mid f_1 \rangle_B = f_2^{\dagger} f_1 \in \text{End}(L^2 B_B) = B$

We aim to build up the tools to understand this definition within this post.

Modules

In general, modules can be thought of as a generalization of a vector space, where we let the scalars come from a ring rather than a field. However, what we're really interested in is modules of algebras. Translating to that language we have:

Definition – Left Module.

VecSpace

Suppose A is a unital algebra. A (left) module ${}_A M$ for A is a vector space M equipped with a bilinear map $\triangleright : A \times M \rightarrow M$ satisfying:

1. (associativity) $(a \triangleright (b \triangleright m)) = (a \cdot b) \triangleright m$ for $m \in M$ and $a, b \in A$
2. (unitality) $1 \triangleright m = m$ for $m \in M$

Definition – Right Module.

VecSpace

Suppose A is a unital algebra. A (right) module M_A for A is a vector space M equipped with a bilinear map $\triangleleft : M \times A \rightarrow M$ satisfying:

1. (associativity) $(m \triangleleft a) \triangleleft b = m \triangleleft (a \cdot b)$ for $m \in M$ and $a, b \in A$
2. (unitality) $m \triangleleft 1 = m$ for $m \in M$

Using these let's understand the definition of a bimodule over two algebras.

Definition: A-B Bimodule.

(VecSpace, UnitalMap)

An A-B bimodule M , denoted ${}_A M_B$ is a right B-module, M_B equipped with a unital algebra map $\cdot : A \rightarrow \text{End}(M_B)$

The Algebraic Relative Tensor Product

For modules of an algebra, $A, M_A, {}_A N$, the relative tensor product $M \otimes_A N$ is the quotient of the tensor product $M \otimes N$ subspace generated by $K = \{ m \otimes n - m \otimes an \mid m \in M, a \in A, n \in N \} \subset M \otimes N$. In short, $M \otimes_A N = M \otimes N / K$. Hence, $ma \otimes_A n = m \otimes_A an$. This gives a sense of what the relative tensor product is supposed to accomplish, but to properly understand Connes fusion, we need to understand the relative tensor product of Hilbert bimodules, for which we have to understand the role of the inner product in this discussion.

Algebra-Valued Inner Products & Hilbert C^* -Modules

Definition: A -valued Inner Product.

(InnerProduct)

Suppose X_A is a right module for a unitary algebra A . An A -valued inner product is a map $\langle \cdot | \cdot \rangle_A : X \times X \rightarrow A$ satisfying:

1. (A -linear in the second variable) $\langle \varphi | \zeta_1 a + \zeta_2 \rangle_A = \langle \varphi | \zeta_1 \rangle_A a + \langle \varphi | \zeta_2 \rangle_A$
2. (conjugate-symmetric) $\langle \varphi | \zeta \rangle_A^* = \langle \zeta | \varphi \rangle_A$
3. (positive-definite) $\langle \varphi | \varphi \rangle_A \geq 0$ $\{\text{in } A, \text{ with equality iff } \varphi = 0\}$

We note here that every unital algebra, A , has the trivial A -valued inner product $\langle a | b \rangle_A := a^* b$

Definition: Right Hilbert C^* -Module.

(Module, InnerProduct)

A right module equipped with an C^* algebra valued inner product is a **right Hilbert C^* -module**.

So at this point it seems like we're pretty close to fusion with an inner product. Let's take a left and right Hilbert C^* -module over B and consider the relative tensor product from above $M \otimes_B N = (M \otimes N) / \{ m \otimes n - m \otimes b n \}$ Recall that by definition we know that M_B and ${}_B N$ are equipped with B -valued inner products. Hence, in the tensor product space $M \otimes_B N$ we might hope to define $\langle m_1 \otimes n_1 | m_2 \otimes n_2 \rangle_{\mathbb{C}} := \langle m_1 | m_2 \rangle_B \langle n_1 | n_2 \rangle_B$ but this is only a well-defined function on $M \otimes_B N$ if it respects the quotient relation $mb \otimes n = m \otimes bn$. Here if we replace m_1 with $m_1 b$ on the left slot: $\langle m_1 b \otimes n_1 | m_2 \otimes n_2 \rangle_{\mathbb{C}} = \langle m_1 | m_2 \rangle_B \langle b n_1 | n_2 \rangle_B = \langle m_1 | m_2 \rangle_B \langle n_1 | b^* n_2 \rangle_B$ using the property $\langle mb | m' \rangle_B = b^* \langle m | m' \rangle_B$ of a right B -valued inner product. Replacing n_1 with bn_1 on the right slot instead: $\langle m_1 \otimes bn_1 | m_2 \otimes n_2 \rangle_{\mathbb{C}} = \langle m_1 \otimes n_1 | m_2 \otimes n_2 \rangle_{\mathbb{C}}$ For these two to agree for every choice of m_i, n_i, b , we need $\langle m_1 | m_2 \rangle_B \langle b n_1 | n_2 \rangle_B = \langle m_1 | m_2 \rangle_B \langle n_1 | b^* n_2 \rangle_B$ $\forall b \in B, \forall n_1, n_2 \in N$. In

words: **acting by b** on one slot of the B -valued inner product must equal acting by b^* on the other slot. This is precisely the statement that the left B -action on N consists of operators that admit adjoints with respect to ${}_B\langle \cdot, \cdot \rangle$, with the adjoint of "acting by b " being "acting by b^* ". So adjointability isn't an aesthetic extra we're bolting on — it's forced on us the moment we ask the relative tensor product to carry an inner product. This motivates the next definition.

Definition: Adjointable Operator.

(InnerProduct)

Suppose X_A, Y_A are right modules equipped with A -valued inner products. A right A -linear map $x : X \rightarrow Y$ is **adjointable** if there is a right A -linear map $x^\dagger : Y \rightarrow X$, called an **adjoint**, such that

$$\langle x^\dagger \zeta \rangle_A = \langle \varphi \rangle_A \quad \text{where } \langle \cdot \rangle_A \text{ is the left inner product on } X \text{ and the right one is on } Y.$$
 We write $End(X_A)$ for the algebra of adjointable operators $X \rightarrow X$.

Trick: Realization.

(VecSpace, InnerProduct)

For any right Hilbert C^* -module X_A , the adjointable maps $Hom(A_A \rightarrow X_A)$ themselves form a right Hilbert C^* -module with A -valued inner product $\langle f \rangle_A := f^\dagger \circ g \in End(A_A) = A$. Moreover the map $X_A \rightarrow Hom(A_A \rightarrow X_A)$ given by $\zeta \mapsto |\zeta\rangle_A$, where $|\zeta\rangle_A : a \mapsto \zeta a$, is a unitary isomorphism of Hilbert C^* -modules. This lets us manufacture a genuine A -valued inner product on a space built out of X by comparing against the standard module A_A .

Correspondences

Definition: Correspondence of Unitary Algebras.

(VecSpace, InnerProduct)

Given A, B unitary algebras. An A - B correspondence is an A - B bimodule ${}_A X_B$ equipped with a right B -valued inner product $\langle \cdot | \cdot \rangle_B$ such that $\rho : A \rightarrow End(X_B)$ is a unital $\star$ -algebra map. Thus,

$$\langle a^* \eta \rangle_B = \langle \eta \rangle_B \quad \text{for all } a \in A, \eta, \zeta \in X$$

From this we can introduce the correct definition of relative tensor product.

Definition: Relative Tensor Product of Correspondences. (VecSpace, InnerProduct)

Given A, B, C unitary algebras and ${}_A X_B, {}_B Y_C$ correspondences. The **relative tensor product**, ${}_A X \boxtimes_B Y_C$, is the quotient space $X \otimes Y / N$ where $N = \text{span}\{\eta \otimes b \otimes \zeta - \eta \otimes b \otimes \zeta; \eta \in X, b \in B, \zeta \in Y\}$. We denote the image of $\eta \otimes \zeta$ by $\eta \boxtimes \zeta$, and equip $X \boxtimes_B Y$ with the C -valued inner product $\langle \eta_1 \boxtimes \zeta_1, \eta_2 \boxtimes \zeta_2 \rangle_C := \langle \eta_2, \eta_1 \rangle_B \langle \zeta_1, \zeta_2 \rangle_C$. This is well-defined and positive-definite (this is the content of Lemma 3.2.33), so ${}_A X \boxtimes_B Y_C$ is again an A - C correspondence.

From Hilbert Space Modules to Correspondences

Definition: Hilbert Space Module.

HilbertSpace

A right **Hilbert space** B -module H_B is a Hilbert space H equipped with a right B -action, i.e. a unital $\star$ -homomorphism $B^{op} \rightarrow B(H)$. Its $\mathbb{C}$ -valued inner product does **not** restrict to a canonical B -valued inner product (unless $B = \mathbb{C}$); this is the content of Warning 3.3.3.

Trick: Realization via $L^2 B$.

(VecSpace, InnerProduct)

Given a Hilbert space module H_B , apply the realization trick with the standard module $L^2 B$ in place of B . The adjointable maps $\text{Hom}(L^2 B_B \rightarrow H_B)$ form a right Hilbert C^* -module with B -valued inner product

$\langle f, g \rangle_B := f^{\dagger} g \in \text{End}(L^2 B_B) = B$. This converts the Hilbert space module H_B – which had no canonical B -valued inner product – into a genuine B -correspondence, exactly the kind of object the relative tensor product of correspondences can act on.

Connes Fusion

Definition: Connes Fusion (the Space).

HilbertSpace

Given a right B -module H_B and a left B -module ${}_B K$, the **Connes fusion relative tensor product Hilbert space** $H \boxtimes_B K$ is $\text{Hom}(L^2 B_B \rightarrow H_B) \boxtimes_B K$ equipped with the $\mathbb{C}$ -valued inner product $\langle f_1 \boxtimes \eta_1, f_2 \boxtimes \eta_2 \rangle := \langle f_2, f_1 \rangle_B \langle \eta_1, \eta_2 \rangle$

$\langle f_2 \rangle \in \langle f_1 \rangle$ where $\langle f_2 \rangle = \langle f_1 \rangle^{\dagger} \in \langle f_1 \rangle$
 $\text{End}(L^2 B_B) = B$.

Definition: Connes Fusion (the Operation).

Bifunctor

Given module maps $x : H_B \rightarrow M_B$ and $y : {}_B K \rightarrow {}_B N$, Connes fusion produces a map $x \boxtimes y : H \boxtimes_B K \rightarrow M \boxtimes_B N$ determined by $f \boxtimes \eta \mapsto (x \circ f) \boxtimes y \eta$ for $f \in \text{Hom}(L^2 B_B \rightarrow H_B)$ and $\eta \in K$. Here x acts on the realized module $\text{Hom}(L^2 B_B \rightarrow H_B)$ by post-composition $f \mapsto x \circ f$. This makes $\boxtimes_B$ a **bifunctor**: it acts not only on modules but on the maps between them, and respects composition.

Previous

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